FMS PEOPLE DEVELOPMENT ROADMAP

From Implementation to Maturity: A Strategic Staging Guide for Maximizing Fleet Management System ROI through Competency.

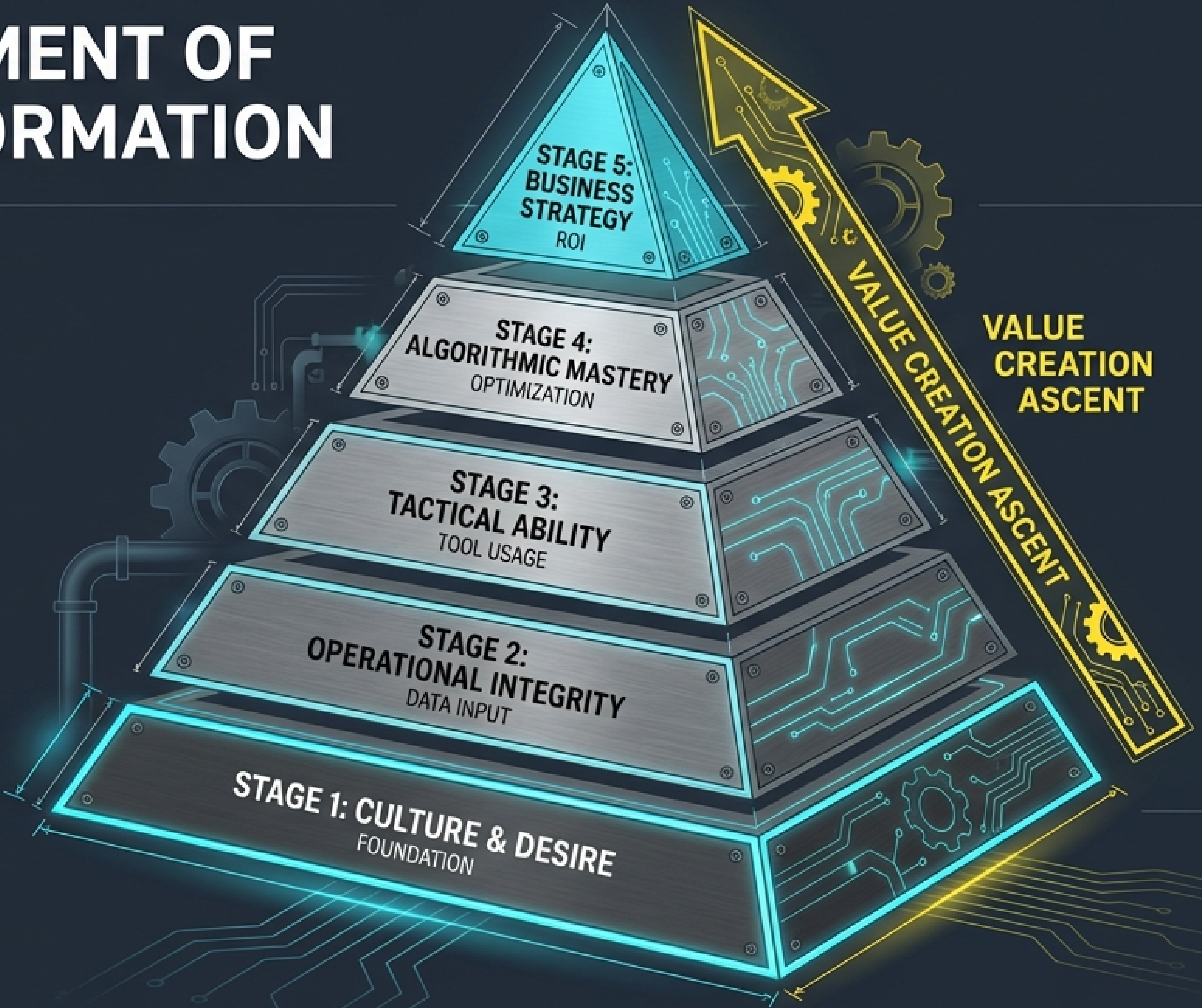
Internal Strategy Document | Operational Excellence Series

THE HUMAN ELEMENT OF DIGITAL TRANSFORMATION

FMS implementation is often mistaken for a software installation. In reality, it is a people development project.

Without the psychological buy-in and technical capability of the workforce, the system remains an expensive tracking tool rather than an optimization engine.

FMS success relies on the symbiotic relationship between Algorithmic Logic and Human Competency.



THE PATH TO OPERATIONAL EXCELLENCE

STAGE 1: FOUNDATION – BUILDING THE CULTURAL BEDROCK

Moving from 'Surveillance' to 'Support'

THE CHALLENGE

- **Resistance:** Addressing fears of "loss of control" or "distrust in technology".
- **Change Management:** Assessment of controller and operator readiness.

THE STRATEGY

- **Campaigns:** Communicating the "Why"—benefits in utilization and cost per ton reduction.
- **Articulation:** Clearly defining "What's in it for me?". The system assists workflow; it does not replace judgment.

STAGE 2: BASIC LEVEL – SECURING DATA INTEGRITY

The Controller as the Gatekeeper of Data Quality

Status Management: Accurate logging of Ready, Down, Delay, and Standby states is non-negotiable.

Validation: Controller validates load-dump logic to prevent "Garbage In, Garbage Out".

Controller Competencies:

- Interface Mastery of HaulRoute & MineGraphics.

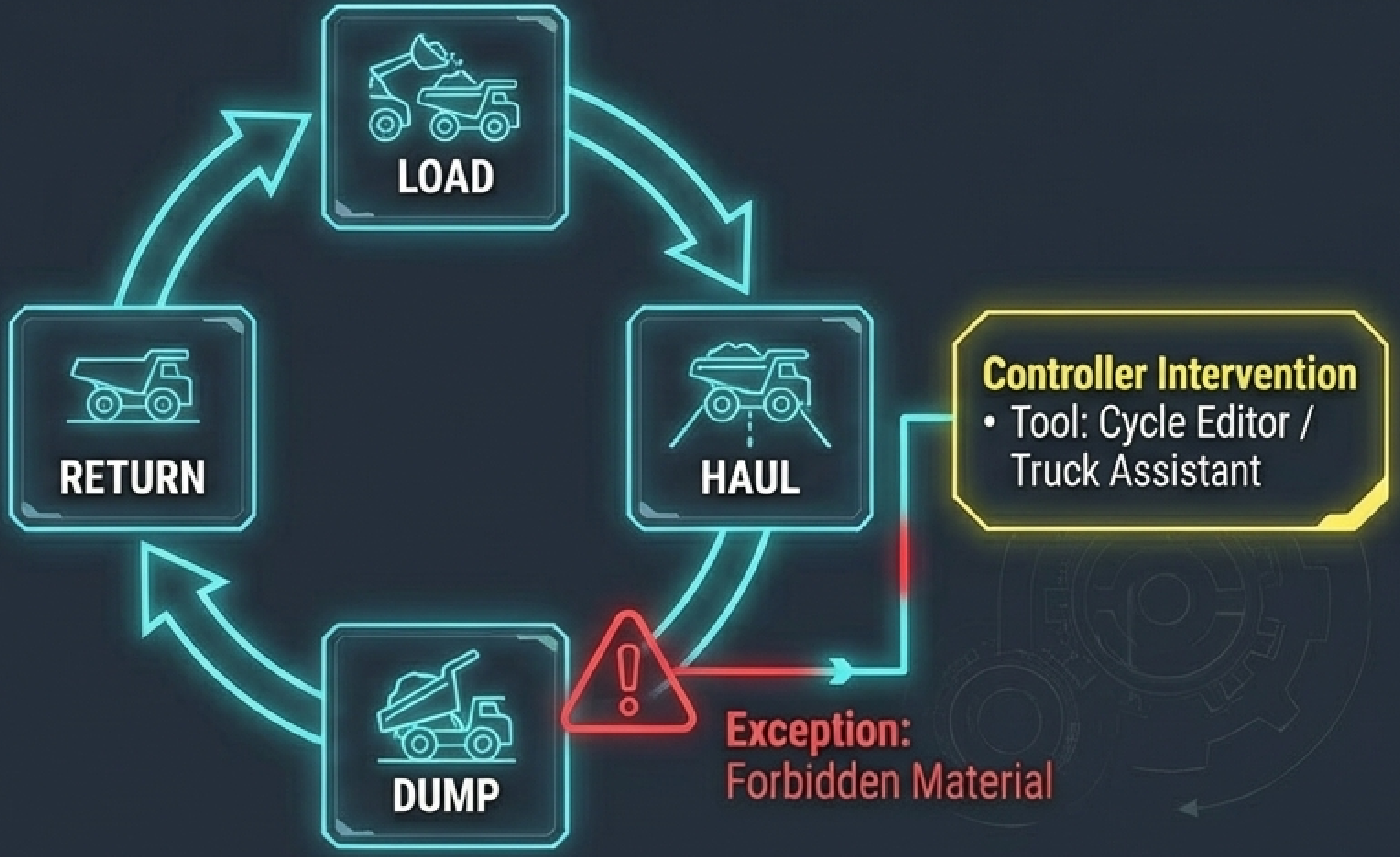
Operator Competencies:

- Digital Discipline.
- Immediate execution of Pre-start Checklists upon system logon.

STAGE 3: INTERMEDIATE – FROM MONITORING TO MANAGEMENT

Utilizing Utilities & Troubleshooting Exceptions

- **Assistant Tools:** Competence in using Truck Assistant, Onboard File Assistant, and Cycle Editor.
- **Triggers:** Understanding Assignment Triggers (e.g., dumping complete prompts reassignment).
- **Troubleshooting:** Solving basic failures in real-time to maintain flow.



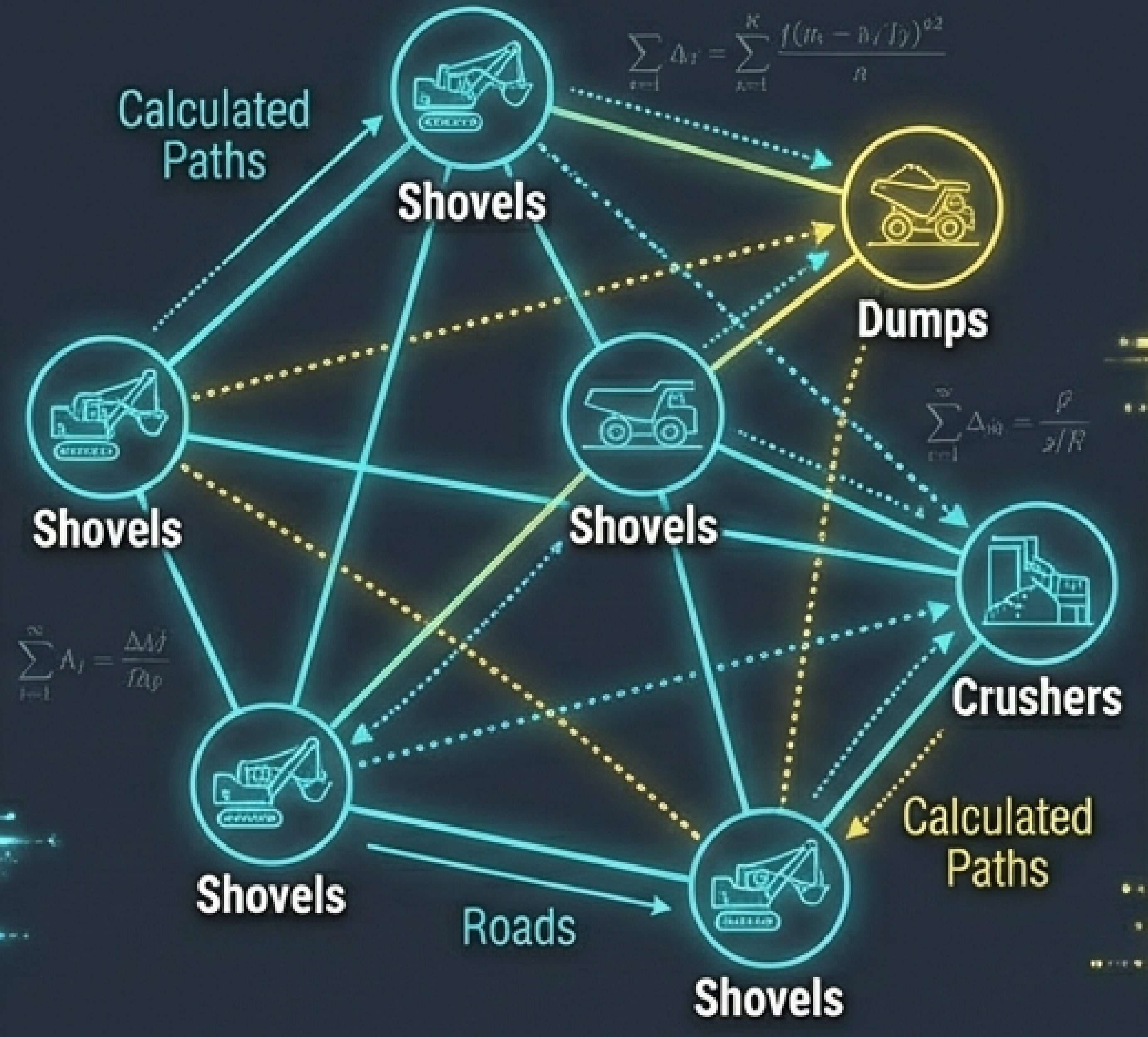
NotebookLM

STAGE 4: ADVANCED – ORCHESTRATING THE MATHEMATICS OF MINING

Algorithmic Optimization for Builders & Senior Controllers

THE LOGIC TRINITY

- **1. Best Path (BP):** The most efficient physical route.
- **2. Linear Programming (LP):** The Master Plan—optimizing for production targets.
- **3. Dynamic Programming (DP):** The Execution—real-time synchronization of Load and Haul.



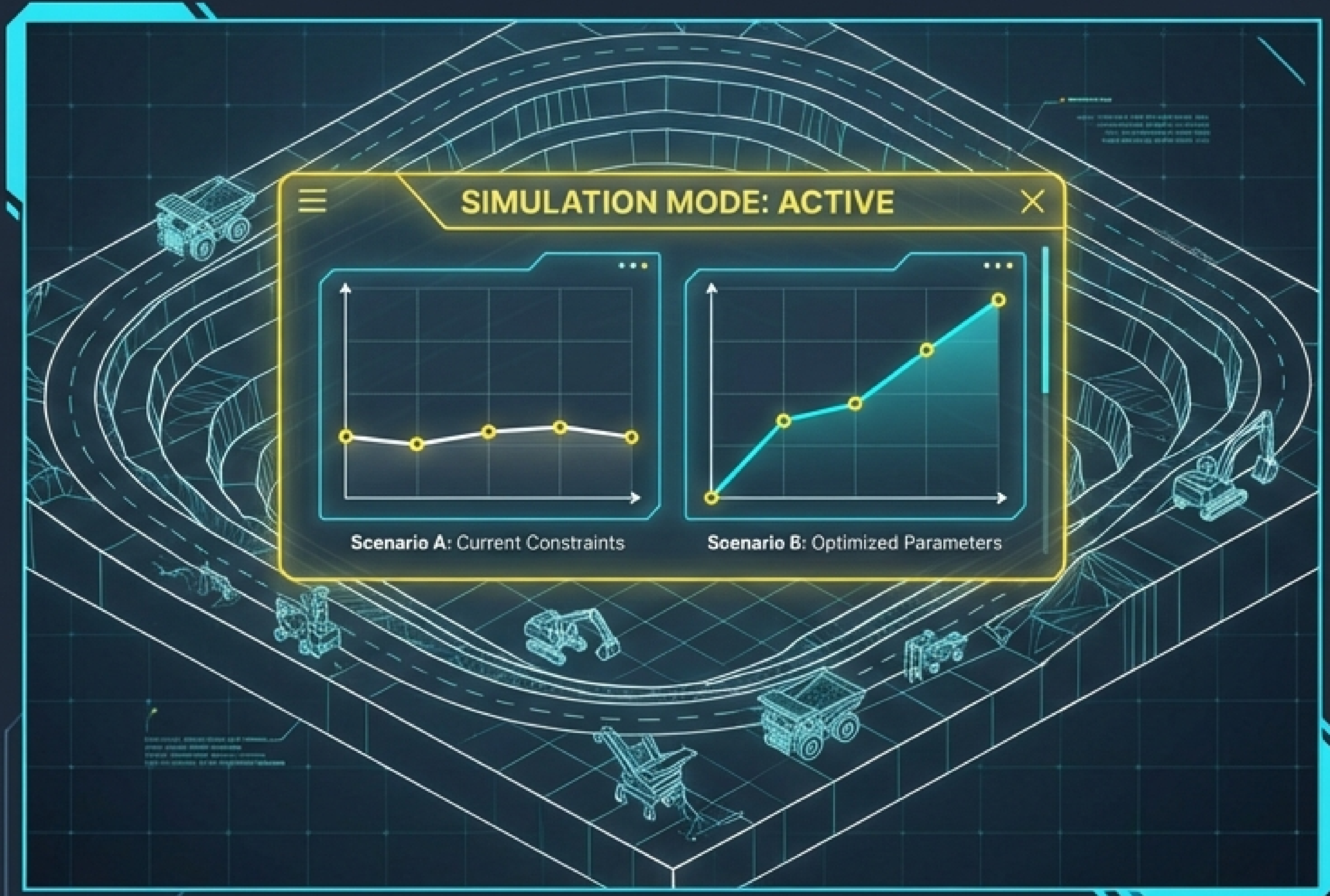
KEY ACTIONS

- **Global Parameters:** Configuring "proportional feeding" for fair truck distribution.
- **Deep Diagnostics:** Investigating "Unused Shovels" or "Over/Under Trucking" within the LP solution.

ADVANCED CAPABILITY: SIMULATION & DECISION SUPPORT

The ‘Sandbox’ Approach: Testing Before Execution

- **The Concept:** Using Decision Support tools to run simulations before altering the live environment.
- **Application:** Evaluating changes in production targets or operational constraints.
- **Result:** Predicting outcomes without risking actual production tonnage.



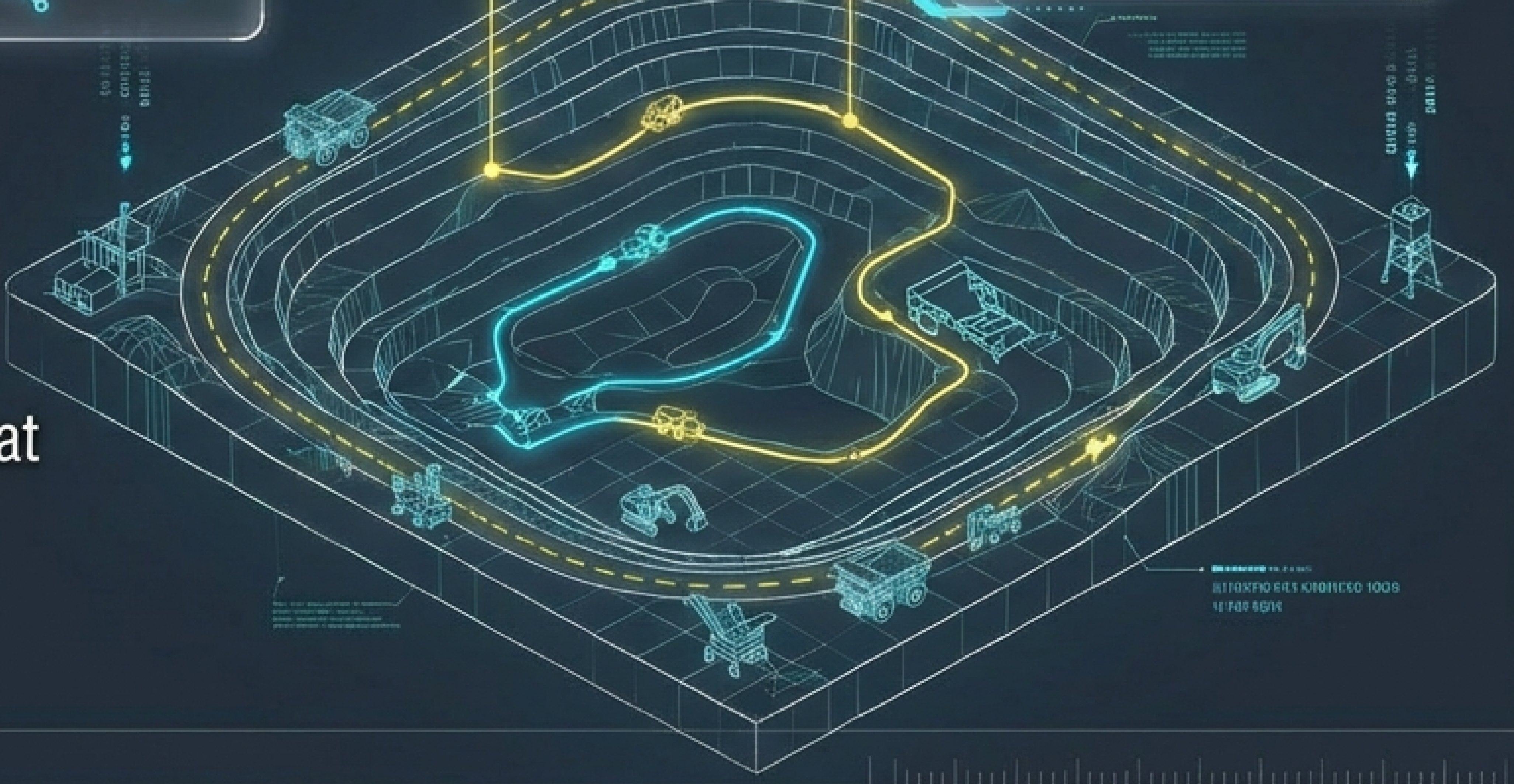
STAGE 5: PROFESSIONAL – STRATEGIC BUSINESS INTEGRATION

Focus on ROI, KPIs, and Continuous Improvement

Production Efficiency Index: **98.4%**

Cost Per Ton: Trending Down

System ROI: **Verified**



- **Controller as Partner:** Moving from a 'dispatcher' mindset to a strategic partner for Production and Maintenance.
- **Data as Asset:** Understanding that a single incorrect elevation point degrades the entire optimization algorithm.

ROLE EVOLUTION: THE MINE CONTROLLER

NOVICE (Stages 1-2)

Focus: Data Entry. Mastering MineGraphics. Validating status updates.

COMPETENT (Stage 3)

Focus: Traffic Management. Handling exceptions, using Truck Assistant, managing fuel logic.

EXPERT (Stages 4-5)

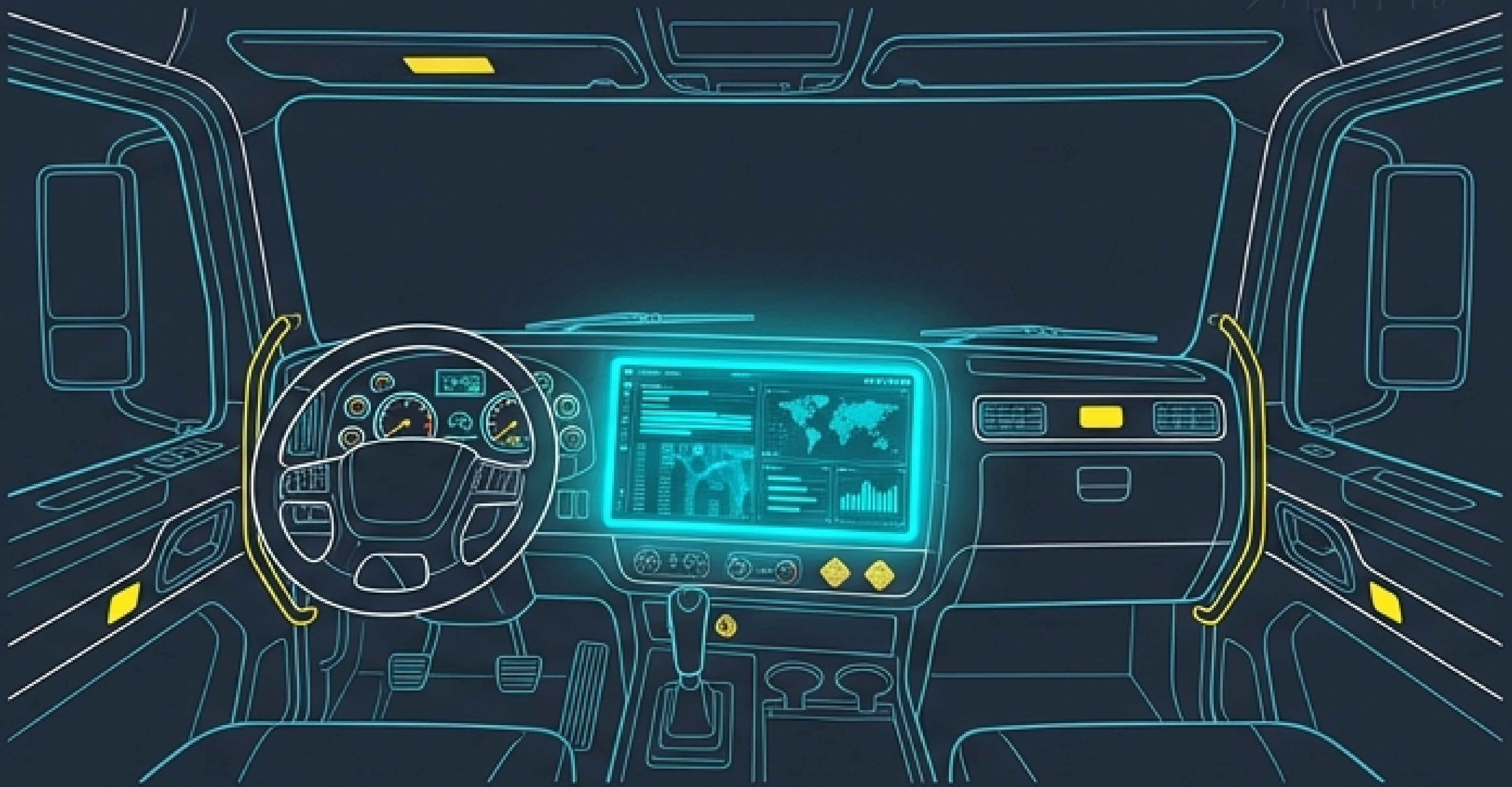
Focus: Strategy. Tuning LP/DP algorithms, configuring global parameters, advising on production flow.

Stage 1-2 | Stage 3 | Stage 4-5

ROLE EVOLUTION: THE CONNECTED OPERATOR

The Shift

- **From:** Intuition-based driving and manual radio communication.
- **To:** Digital discipline. Responsiveness to auto-assignments and trust in the “Black Box” instructions.



The operator’s compliance with the system is the “pulse” that keeps the algorithm accurate. Digital checklists (Pre-start) are the first line of defense.

MECHANICS OF SUSTAINMENT: TRUST BUT VERIFY

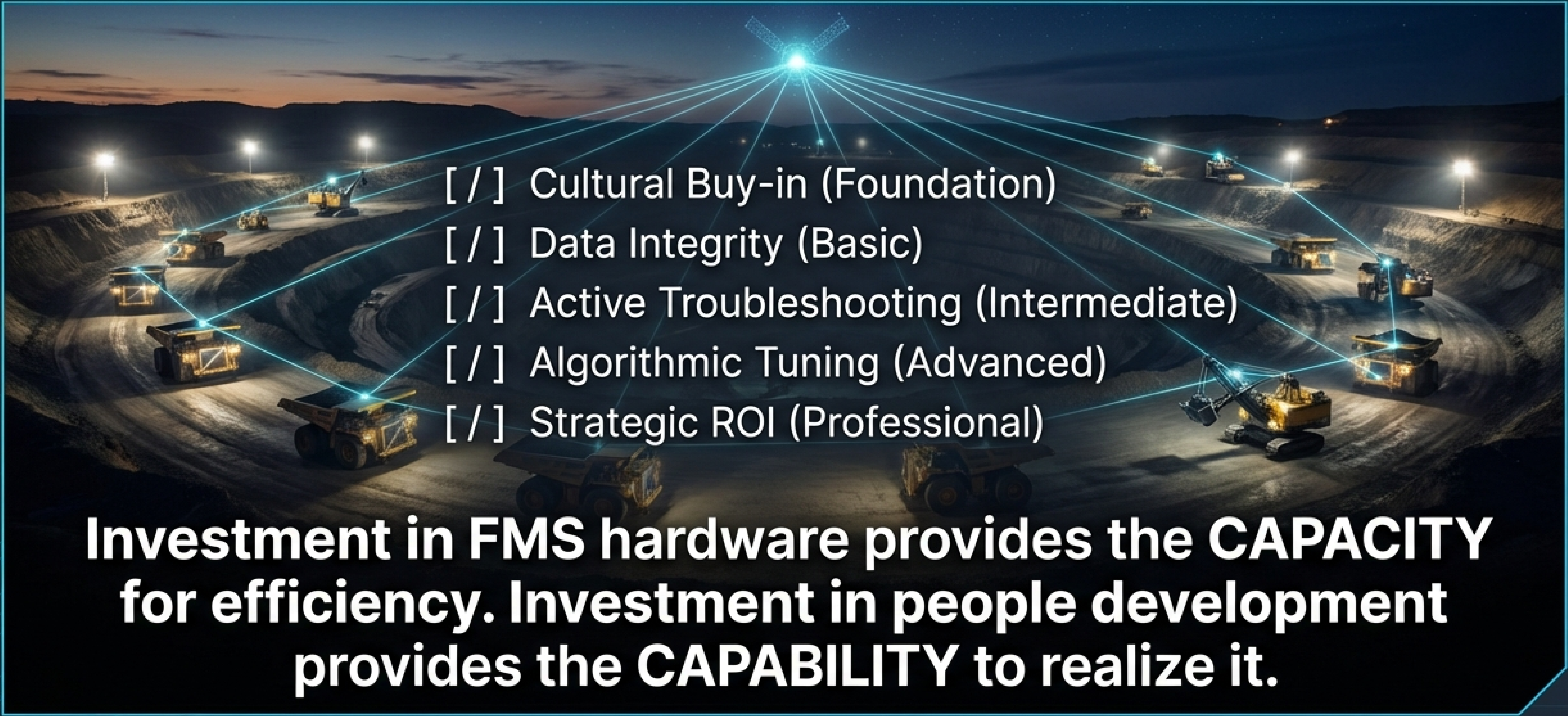
Preventing Competency Decay

Identify gaps for retraining.

Live execution.

Continuous verification ensures that the ‘Data Integrity’ foundation (Stage 2) remains solid year-over-year. Competency audits prevent the slide back into manual habits.

ACHIEVING TOTAL FLEET MANAGEMENT MASTERY



[/] Cultural Buy-in (Foundation)

[/] Data Integrity (Basic)

[/] Active Troubleshooting (Intermediate)

[/] Algorithmic Tuning (Advanced)

[/] Strategic ROI (Professional)

Investment in FMS hardware provides the CAPACITY for efficiency. Investment in people development provides the CAPABILITY to realize it.

Ready for Tomorrow.